

TEJASWI ANAND

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Education

Faculty of Technology, University of Delhi

Aug '24 – July '28

B.Tech in Computer Science & Engineering (Blockchain, Cybersecurity & Cryptography)

CGPA: 8.18

Coursework: Data Structures, Algorithm Analysis, Database Management, Operating Systems, Cloud Infrastructure

Experience

Software Engineer Intern – Nowlez: AI Munshi

July '26 – Present

Python, LangGraph, LangChain, Gemini, JavaScript, HTML, CSS

- Engineered an **ephemeral sandbox backend** enabling the AI assistant to autonomously execute scripts to generate and edit documents using **Gemini models**, capping REPL history to prevent memory growth.
- Architected an **asynchronous research subagent** ('deep_explore') executing long-running tasks on isolated worker slots with **webhook wake-ups**, ensuring zero UI blocking during concurrent multi-turn chats.
- Built core delivery infrastructure by connecting a self-deployed **LangGraph Agent Server** to the client via webhooks for **Human-in-the-Loop (HITL)** confirmations and voice transcriptions.
- Designed a **multi-IP request-routing infrastructure** on a VPS, distributing e-Courts API requests across **6 public IPs** to overcome a **0.33 req/sec** limit and raise throughput by **6×**.

Projects

AKTU DAA RAG System | *Python, FastAPI, LangChain, GLM-5.3, BM25, WeasyPrint*

Oct '26

- Architected a **production-grade RAG system** with a dual-engine classifier (LLM router + lexical regex) to accurately disambiguate 29 syllabus topics and eliminate cross-unit keyword collisions.
- Implemented a **Dual Hybrid Retrieval pipeline** extracting top-20 candidates via dense semantic search (BAAI/bge-base embeddings) and BM25, fused via **Reciprocal Rank Fusion (RRF, k=60)**.
- Designed a **context-aware chunking algorithm** utilizing **LangChain and Regex** to process study materials into **900-character windows**; developed an intelligent metadata-tagging system anchoring texts to parent topics, mitigating context-loss and boosting vector search precision.
- Integrated an out-of-scope similarity floor gate (cosine thresholds: 0.45/0.58) to block hallucinations, paired with **Server-Sent Events (SSE)** for streaming and **LangSmith** for deep trajectory telemetry.

DeepFake Audio Detection Pipeline | *Python, FastAPI, PyTorch, Hugging Face Transformers*

May '26

- Built a **WhatsApp bot in an 8-hour hackathon** that checks voice notes for AI-generated speech and replies Real / Suspicious / Fake with a score, at **~1.3 s of model time per 10-second clip**.
- Developed a **3-signal detection ensemble** combining AASIST (a 298K-parameter graph-attention network), **wav2vec2 embedding analysis**, and 5 audio-metadata heuristics, with automatic GPU-to-CPU fallback.
- Designed an event-driven backend where a FastAPI webhook feeds a **Redis queue drained by Celery workers**, secured with HMAC-SHA256 signature checks and backed by **41 pytest tests**.

Project Eligify | *Python, Flask, SQLAlchemy, Tesseract OCR, OpenCV* | [GitHub](#)

Nov '25

- Engineered a **hybrid PDF + OCR extraction pipeline**: PyPDF2 with a Tesseract fallback running **20 OCR passes per image over 5 OpenCV-preprocessed versions**, reading DigiLocker PDFs and phone photos.

Research Experience

Summer Research Intern – SoCSE, RV University

June '26 – July '26

Secure Routing Protocol for IoT Networks using Lightweight Cryptography | Python, NS3, Cooja

- Built **SecureRPL**, a lightweight security extension of the IETF RPL (RFC 6550) IoT routing protocol in Python, combining **ChaCha20-Poly1305 AEAD**, replay protection, trust-based routing, and key revocation.
- Engineered a reproducible benchmarking framework (11 Python modules, ~1.5K LOC) running **330 seeded simulations (132K packets)** across 4 topologies in **under 3 seconds**.

Technical Skills

Languages: Python, JavaScript, SQL, C, HTML/CSS, PowerShell

AI/ML: LangChain, LangGraph, Transformers, LLM APIs (Gemini, GLM-5.3), PyTorch, Hugging Face

Backend & Architecture: FastAPI, Flask, Agent Servers, RAG, REST APIs, Celery, Redis, Supabase

Cloud & Tools: Docker, Git, GitHub, AWS, DigitalOcean, Cloudflare R2, Linux, WSL, LangSmith

Achievements & Leadership

- 3rd Place among 50+ teams**, AI War (IEEE Student Branch, University of Delhi) – 8-hour AI competition (May '26)
- TEDxFOT Licensee & Organizer** – selected by TED; leading a multidisciplinary team (Sept '25 – Sept '26)